CSI 5430 Assignment1

Name: Lingjia Liu

Student Number: [redacted]

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This report covers some explorations about fitting, generalizing and regularization of polynomial regression models. This model implements GD to estimate coefficients, plots figures to present variation in MSE by varying three factors, including model complexity (d), sample size (N) and noise level (σ), show the differences between using and not using weight decay. According to the results, I compare and then make conclusions. All the .py documents, MSE data and pictures have been packaged together.

The relevant Parameter setting is: .

Numbers of trails(M): 50

Learning rate: 0.01

Epochs:1000

Weight decay: 0.1

Model complexity: d∈ {0,1,2…,20}

Sample size: N ∈ {2,5,10,20,50,100,200}

Noise level: σ ∈ {0.01, 0.1, 1}

To be noticed, the three factors: model complexity (d), sample size (N) and noise level (σ) will not vary at the same time.

Sample size

First, under fixed model complexity and noise level, we observe the impact of sample size on MSE (mean squared error). Then, we compare the differences of MSE in different model complexity side-by-side. Finally, we find the changes on MSE of those models after utilizing weight decay regularization.

Sample size: N ∈ {2,5,10,20,50,100,200}

noise level/variance/sigma: σ=0.1

model complexity: d= [3,10,17]

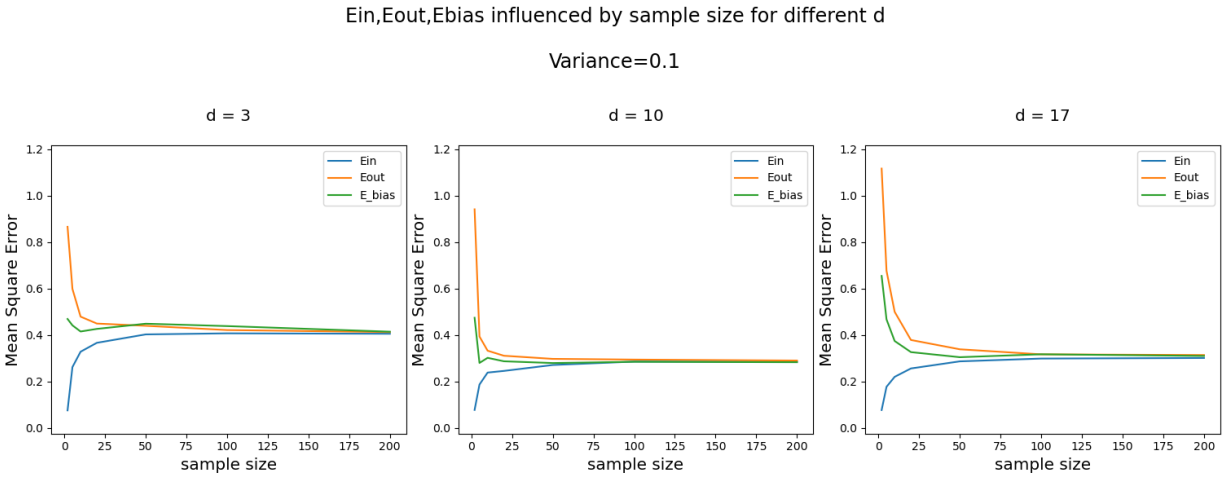


Figure 1: impact of sample size -without regularization

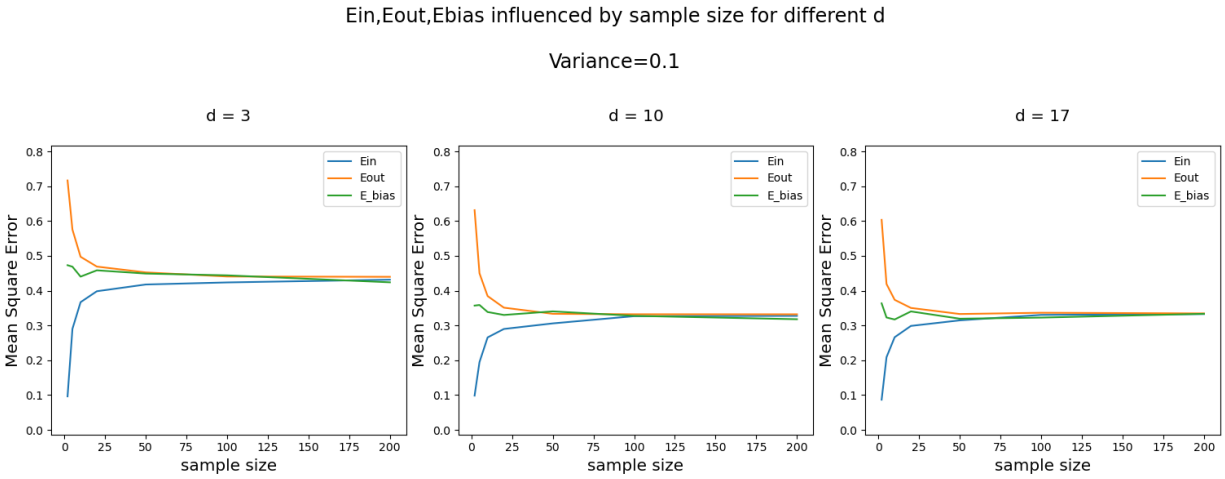


Figure 2: impact of sample size -with regularization

Figure1 and Fingure2 similarly shows that, as the sample size increases, E_{out} decreases dramatically and it is true for E_{bias} at an early stage, whereas E_{in} shows an opposite trend. In the following, E_{in} E_{out} E_{bias} begin to converge and gradually tend to stable eventually.

By comparing different complexities, it can be found that E_{gen} (E_{out}-E_{in}) of the complex model (d=17) is significantly higher than that of the simple model(d=3) when the sample size is less than 10 (N<10). Hence, in order to perform a better accuracy, the sample size of complex model is recommended to be large. In addition, increasing complexity (from d=3 to d=17) can reduce MSE (from 0.4 to 0.3) obviously.

Comparing the regularized model with the non-regularized model (Figures 1 and 2), the former has lower maximum values of both E_{out} and E_{bias} at the beginning (N=2,5). Furthermore, E_{gen} (E_{out}-E_{in}) which reduces by nearly 0 in the former is earlier than that in the latter. This is obvious in model(d=17), the regularized model (N=75) presents almost same E_{gen} with non-regularized one(N=100). Therefore, regularization can be treated as 'soft' enlarging the number of data set.

Model complexity

First, under fixed sample size and noise level, we find the effect of sample size on MSE (mean squared error). Then, we compare the changes of MSE by varying sample size (side-by-side). Finally, we make comparisons on MSE of those models after utilizing weight decay regularization.

Sample size: N = [20,100]

noise level/variance/sigma: σ=0.1

model complexity: d ∈ {0,1,2…,20}

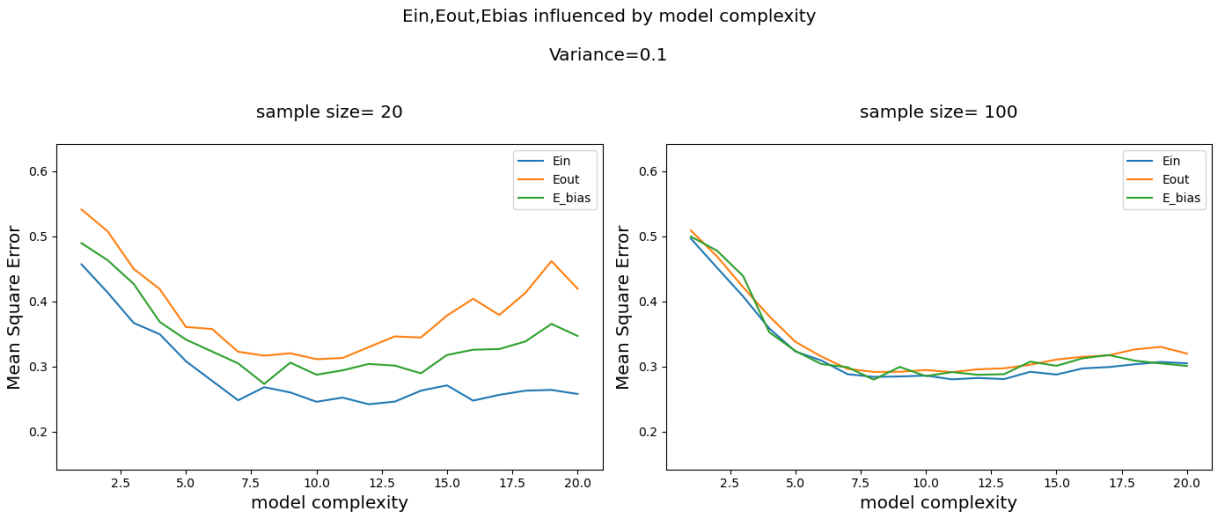


Figure 3: impact of model complexity-without regularization

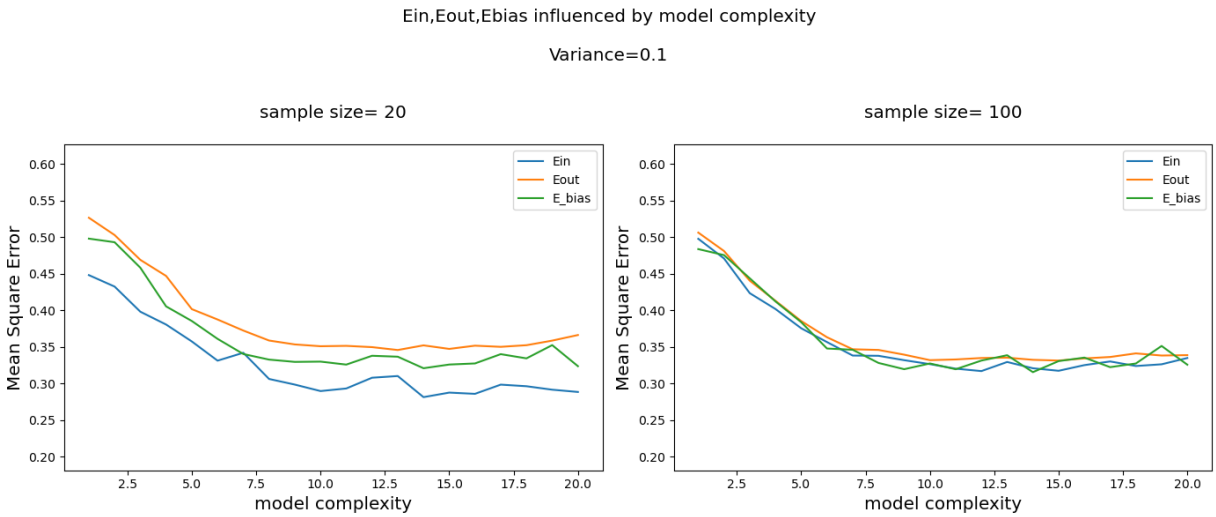


Figure 4: impact of model complexity-with regularization

Figure3 and Fingure4 similarly shows that, with model complexity goes up, E_{out} and E_{in} gradually decrease before keeping steady. Strikingly, in the left chart in Figure3, after decreasing, E_{out} reach the bottom which divide the model into the under-fitting model and over-fitting model, subsequently it increases, while E_{in} declined gradually, leading to the increase in E_{gen} (E_{in}-E_{out}). It presents the optimal complexity(d=9), dividing the under-fitting and over-fitting models.

By comparing different sample size, E_{out} of complex model (d=18) shows lower MSE when the sample size is larger (N=20 VS N=100), which means increasing in sample size can reduce over-fitting efficiently.

Compared regularized model with non-regularized one (Figure3 and Figure4), it can be observed that regularization can alleviate over-fitting. This is extremely apparent when sample size equals to 20(N=20), E_{out} of complex models with regularization experience significant declines from that without regularization, to be specific, from over 0.4 to almost 0.35. Furthermore, compared regularized model (sample size =100) with non-regularized one, it shows that the former can bring earlier efficient outcomes E_{out}(d=10) than the latter(d=11) on the same data set, which can be treated as ‘soft’ reducing model complexity.

Noise level/Variance

First, under fixed sample size and model complexity, we observe variation of MSE (mean squared error) caused by noise level. Then, we compare the MSE of those models after utilizing weight decay regularization, analyse unusual results and then change one of three parameters to present a clear explanation.

Sample size: N = 100,200

noise level/variance/sigma: $\sigma \in \{0.01, 0.1, 1\}$

model complexity: d = 8

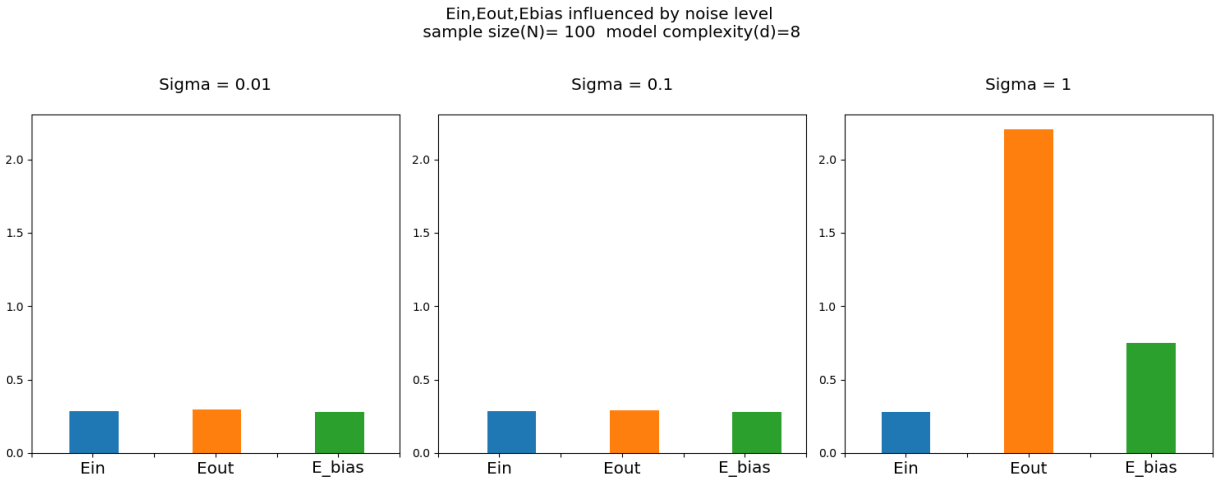


Figure 5: impact of noise level-without regularization(N=100,d=8)

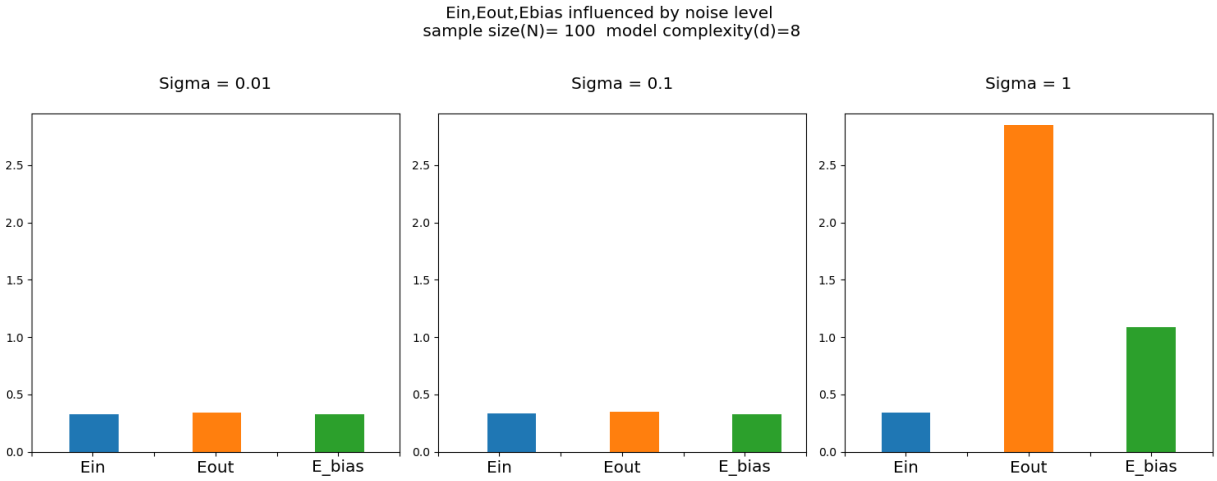


Figure 6: impact of noise level-with regularization

The common in Figure5 and Figure6 is that, with the rise of variance, E_{out} E_{bias} show sharpen climb, whereas E_{in} keep almost stable. In Figure5, the given sample size N is 100 and model complexity d is 8, E_{in} E_{out} E_{bias} are in the same quantity at start (around 0.35). As variance increasing from 0.01 to 1, E_{out} E_{bias} increase obviously to 2.1 and 0.9 respectively. Hence, when in order to enhance the accuracy of complex model with small sample size, data should be 'clean' without too much noise level (lower variance).

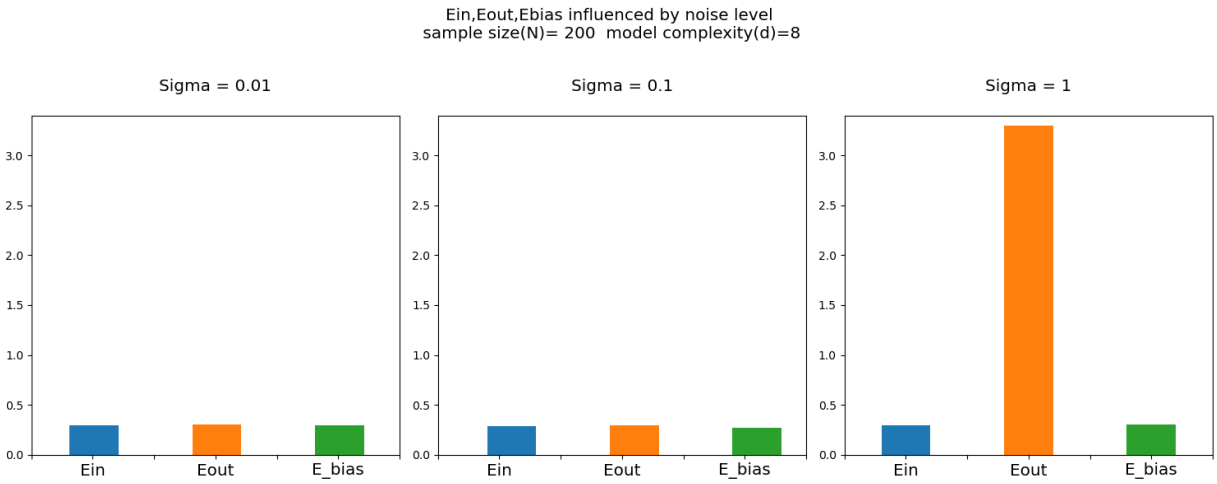


Figure 7: impact of noise level-without regularization

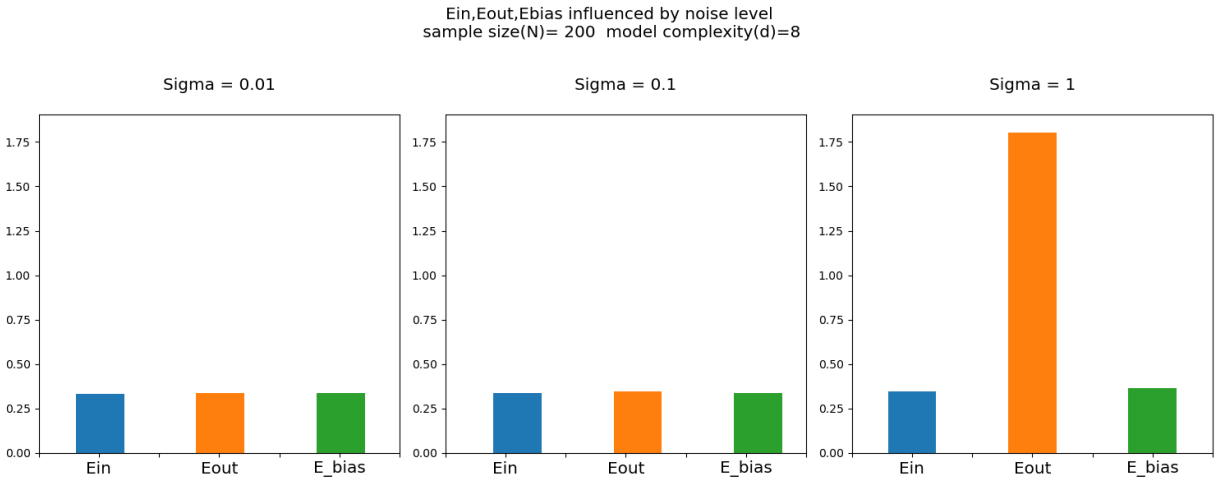


Figure 8: impact of noise level-without regularization

Compared Figure5 with Figure6, it is surprised that gap between E_{in} and E_{out} slightly increase after regularizing (particularly sigma=1), which attribute to small sample size.

Trying to enlarge amount of data set to 200, shown in Figure7 and Figure8, the gap apparently shrinks. So, when noise at a high level, a rise in sample size can also improve model performance.